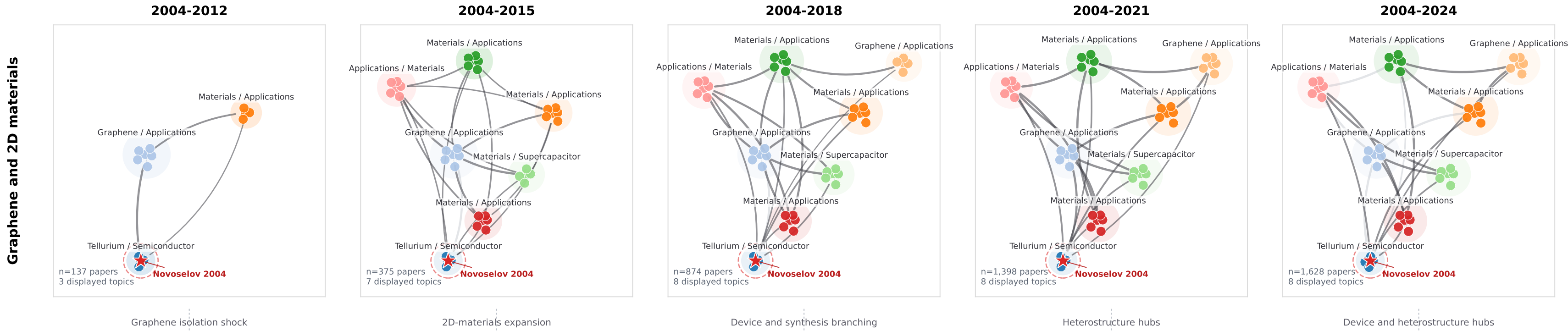


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

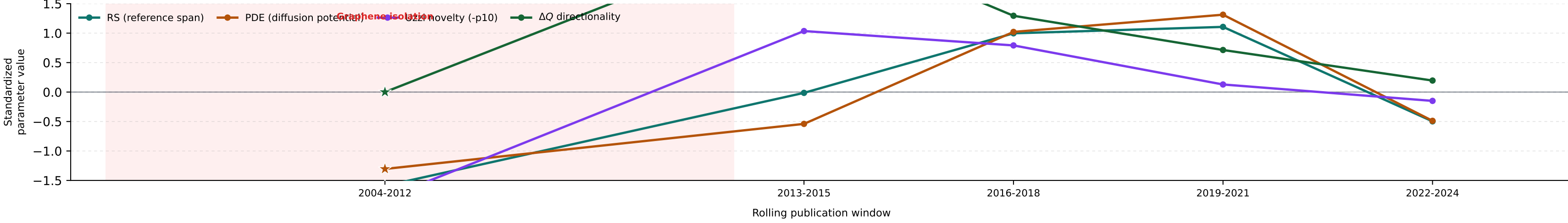
Example domain: Graphene and 2D materials. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



b Dominant parameter trajectories explaining the graph transitions in panel a



Expansion
RS up + PDE up
Physics, materials and devices connect

Novelty
Uzzi up
Atypical recombination of carbon physics

Field formation
DeltaQ later > 0
Stable 2D-materials subfield emerges